

Part 1

건식전극 제작

활물질 비율, 로딩

■ 활물질 비율, 로딩 (No.1, No.2, Poly(811) 동일)

- AM:SE:VGCF:PTFE = 80:18:1:1
- No.1 비용량 205.9mAh/g, No.2 비용량 213mAh/g, Poly 비용량 200mAh/g
면용량 3기준 로딩 각각 14.57, 14.08, 15 mg/cm²

Physical & Chemical properties

LF innoxide

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Sample			No.1	No.2
Main Elements	Ni/(Ni+Co+Mn)	mol%	82.5	86.9
	Co/(Ni+Co+Mn)	mol%	12.7	5.7
	Mn/(Ni+Co+Mn)	mol%	4.8	7.4
	Li/(Ni+Co+Mn)	mol%	0.99	1.02
PSD	D50	μm	3.94	3.84
	D10		2.84	2.55
	D90		5.56	5.98
	Dmin		2.12	1.50
	Dmax		9.25	11.00
BET		m ² /g	0.56	0.57
Tap Density		g/cc	2.14	2.06
pH		-	11.31	11.38
Excess Li ₂ CO ₃		%	0.42	0.19
Excess LiOH		%	0.07	0.07
Moisture		ppm	127	188

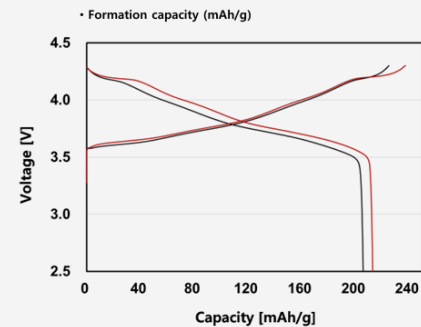
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전지 평가 결과

LF innoxide

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Item	Coin cell 평가 (2.5-4.3V, 25°C)			
	0.1C Formation			
	CC	DC	Eff	DCIR
	[mAh/g]		[%]	[Ω]
No.1	225.0	205.9	91.5	23.5
No.2	237.3	213.0	89.8	27.5

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■ 전극제작 (최신버전)

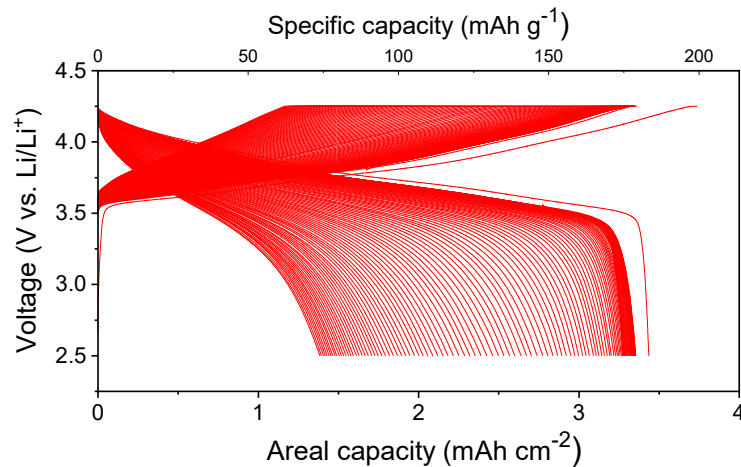
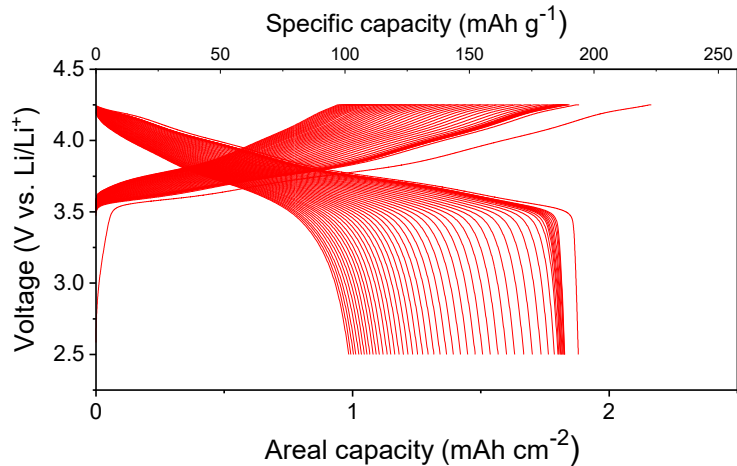
1. 2g base AM/SE/VGCF 각각 1.6g/0.36g/0.02g 넣고 핸드 믹싱 (10분)
2. 큰 컨테이너에 넣고 5분 2000RPM 씩키 믹싱 (zirconia ball 5pi 6개, 3pi 9개)
3. 수득 이후 PTFE 0.01g 넣고 살살 믹싱 이후 볼밀 1시간 200RPM (zirconia ball 5pi 5개, 2pi 15개)
4. 수득 이후 씩키 믹싱하고 PTFE 0.01g 추가로 넣고 씩키 믹싱
5. 수득 이후 핸드 믹싱하여 떡전극 만든 이후 hot rolling 진행
(8방향 * 5단계로 힘 주면서 * 총 5회)
6. 로딩 타겟팅하여 추가 롤링 이후 10pi로 타발

Part 2

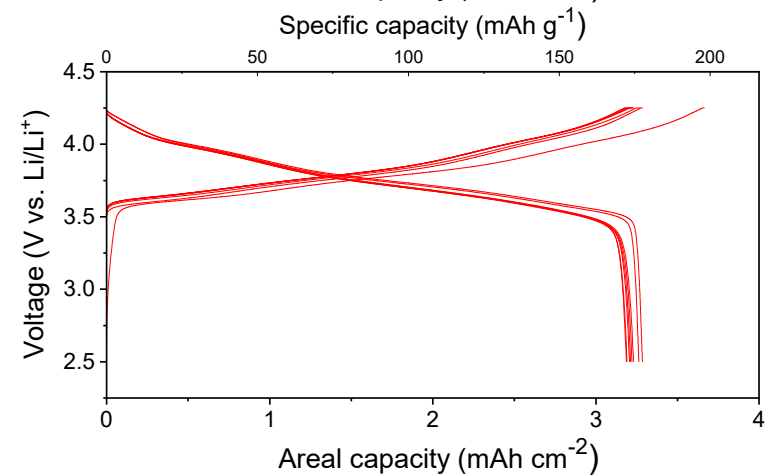
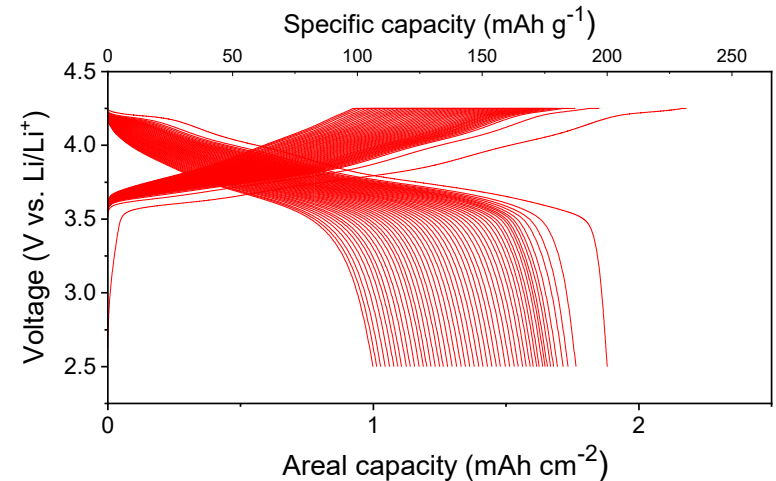
No.1, No.2, Poly 성능평가

No.1, No.2 powder

■ No.1, No.2 profile (전해질 바뀌기 전)



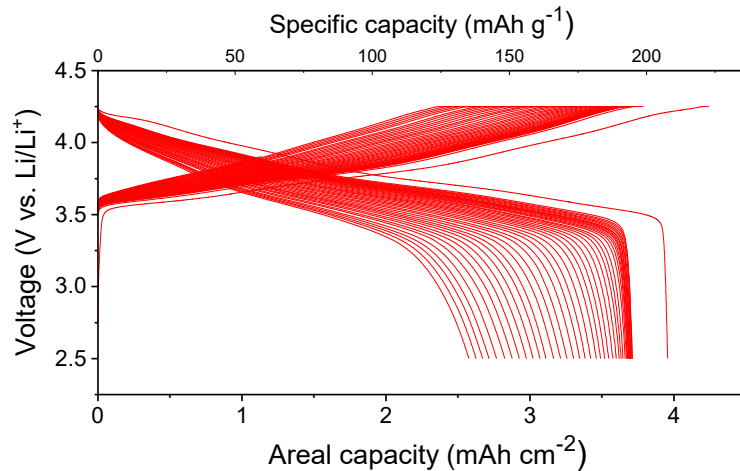
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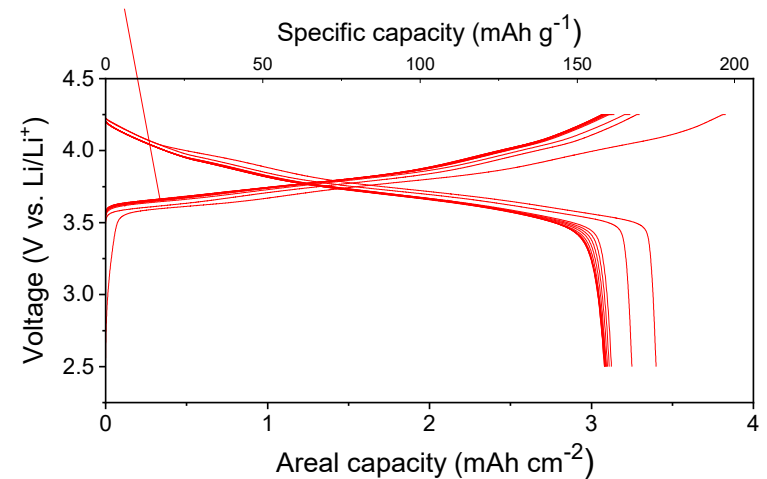
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No.1, No.2 건식전극

■ No.1, No.2 profile (전해질 바뀌기 전)



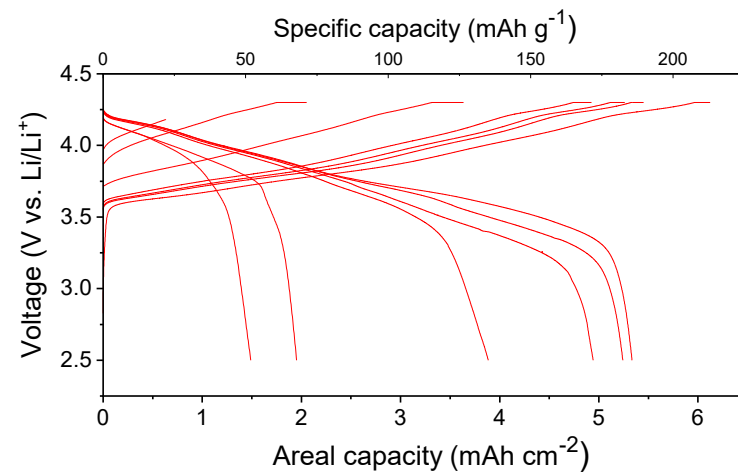
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No.1, No.2 건식전극

■ No.1, No.2 profile (후막)

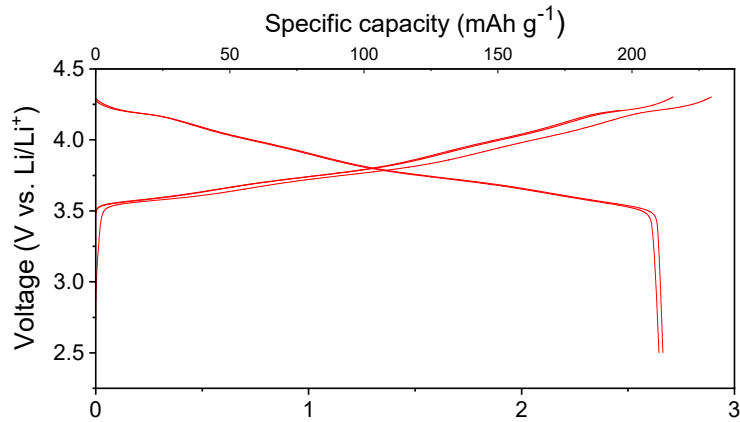


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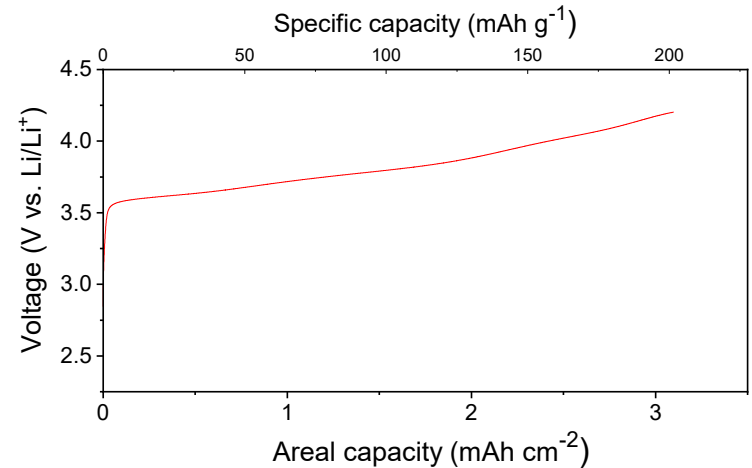
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No.1, No.2 건식전극

■ No.1, No.2 profile (박막)



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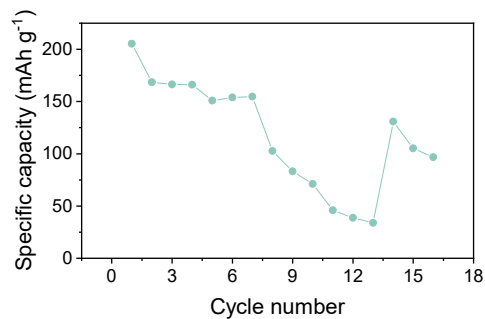


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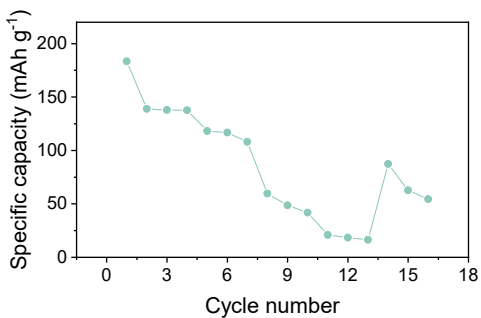
No.1, No.2 건식전극

■ No.1, No.2, poly profile (율속)

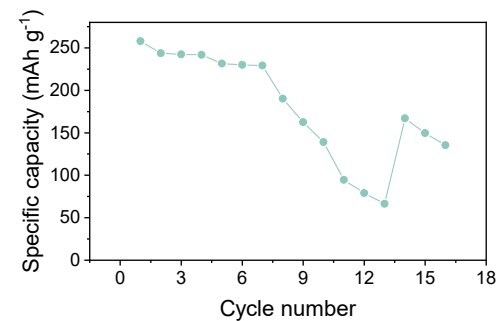
- 0.1C (1cycle) – 0.33C (3cycle) – 0.5C – 1C – 2C - 0.33C



< No.1 dry >



< No.2 dry >



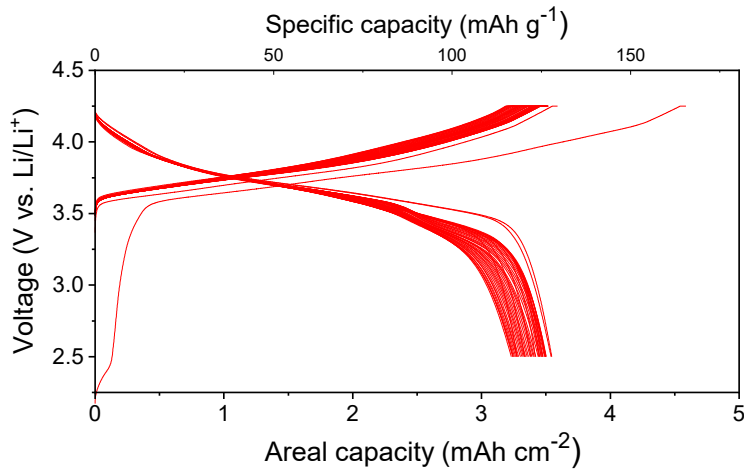
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Part 2

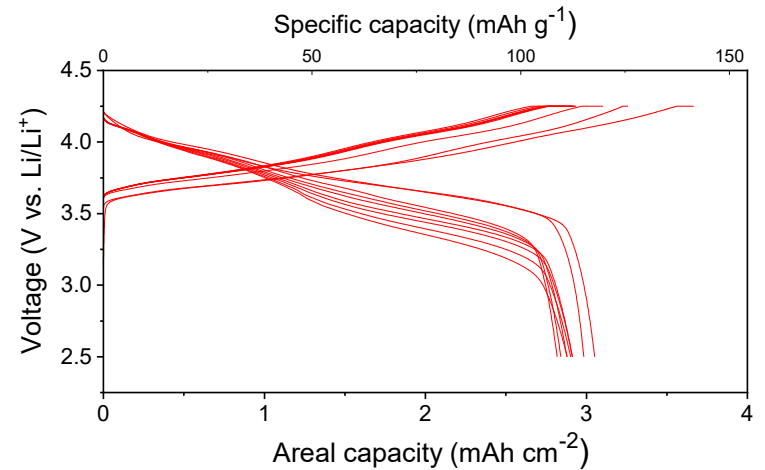
Bimodal 관련

No.1, No.2 Bimodal

■ No.1, No.2 7:3 ($\sim 4 \frac{\text{mAh}}{\text{g}}$)



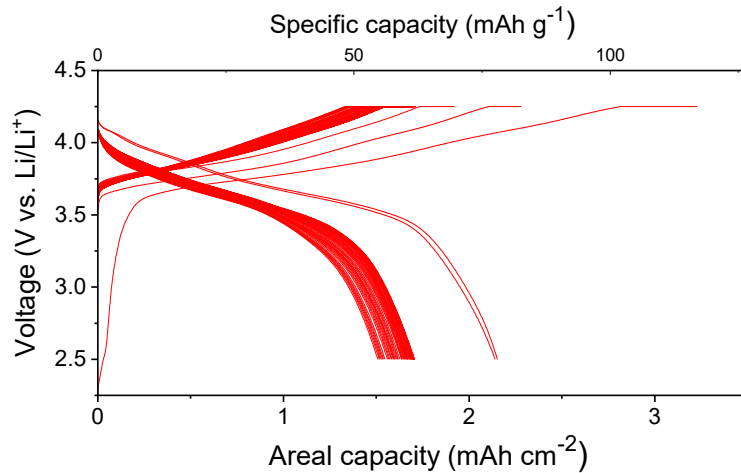
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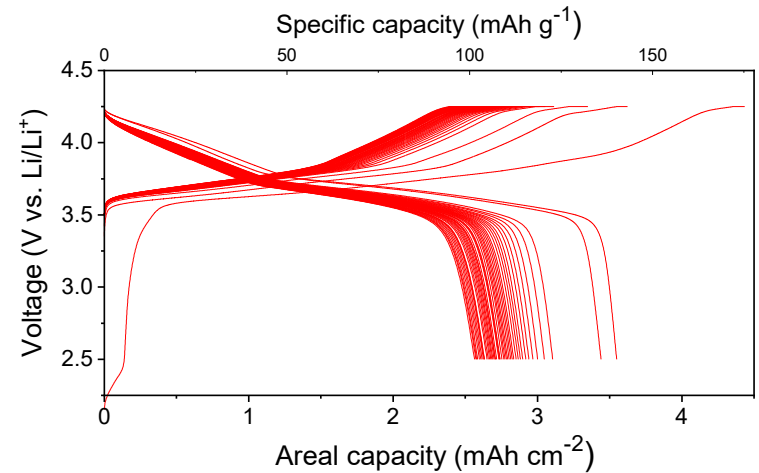
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No.1, No.2 Bimodal

■ No.1, No.2 7:3 (~5월)



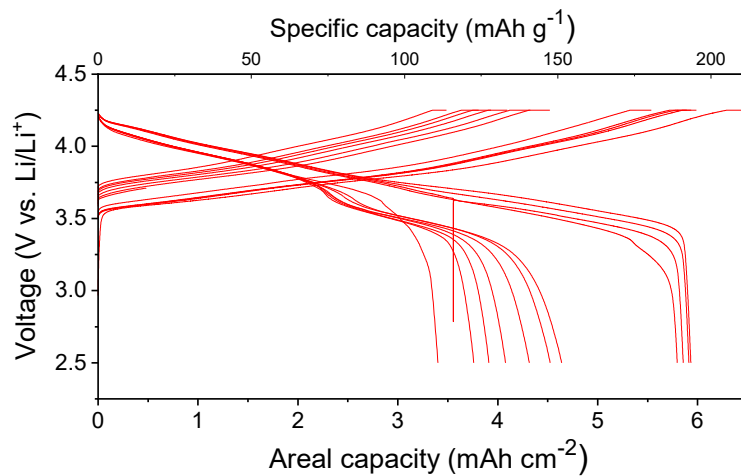
< No.1 7:3 dry >



< No.2 7:3 dry >

No.1, No.2 Bimodal

■ No.1 3:7 ($\sim 6 \frac{\text{wt}}{\text{wt}}$)



< No.1 3:7 dry >